

opportunity on the Sun, and by evaluating weather conditions on Earth and recommending Earth Resources opportunities, crewmen were instrumental in attaining extremely high quality solar and Earth oriented data.

Upon completion of the engineering tests, Skylab was positioned into a stable attitude and systems were shut down. It was expected that Skylab would remain in orbit eight to ten years. However, in the fall of 1977, it was determined that Skylab was no longer in a stable attitude as a result of greater than predicted solar activity.

The empty Skylab spacecraft returned to Earth July 11, 1979 scattering debris over the Indian Ocean and the sparsely settled region of Western Australia.

Space Shuttle

After the Apollo-Soyuz mission, the increasing cooperation of space efforts between the two space programmes led to rapid advances in discovery. While the Skylab was retired in 1979, NASA still had access to the Salyut space stations and was working with the European Space Agency on the Soviet Space Programme's upcoming Mir space station.

One of the major technological breakthroughs came at the time of the retirement of the Skylab, when NASA perfected its space shuttle fleet of aircraft.

The space shuttle is the world's first reusable spacecraft, and the first spacecraft in history that can carry large satellites both to and from orbit. The shuttle launches like a rocket, manoeuvres in Earth orbit like a spacecraft and lands like an airplane. Each of the five space shuttle orbiters – Discovery, Atlantis Challenger, Endeavour, Columbia – were designed to fly at least 100 missions. The Enterprise was the only shuttle to be used only for tests and it never made it into space. After testing through the late 70s, the first Space Shuttle Columbia was launched in 1981. This was soon followed by Challenger (1983), Discovery (1984), Atlantis (1985), and Endeavour (1992, built after the Challenger's accident in 1986).

The space shuttle consists of three major components: the Orbiter which houses the crew; a large external fuel tank that holds fuel for the main engines; and two solid rocket boosters which provide most of the shuttle's lift during the first two minutes of flight. All of the components are reused except for the external fuel tank, which burns up in the atmosphere after each launch.

The Orbiter is both the brains and heart of the Space Transportation System (STS), the official title for the space shuttle programme. About the same size and weight as a DC-9 aircraft, the Orbiter contains the pressurized crew compartment (which can normally carry up to seven crew members), the huge cargo bay, and the three main engines mounted on its aft end.

The cockpit, living quarters and experiment operator's station are located